Practical 9

In [62]:

```
from sklearn.datasets import fetch_20newsgroups
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report

# -------- Load Dataset --------
data = fetch_20newsgroups(subset='all')

X = data.data
y = data.target

# -------- Train-Test Split --------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# -------- Feature Extraction --------
vectorizer = TfidfVectorizer(stop_words='english', max_features=10000)

X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

# -------- Train Model --------
model = MultinomialNB()
model.fit(X_train_vec, y_train)

# -------- Prediction --------
y_pred = model.predict(X_test_vec)

# -------- Evaluation --------
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

# -------- Test Custom Input --------
text = ["NASA launched a new satellite into space"]
text_vec = vectorizer.transform(text)

pred = model.predict(text_vec)
print("\nPredicted Category:", data.target_names[pred[0]])
```

Accuracy: 0.8673740053050398

Classification Report:

	precision	recall	f1-score	support
0	0.83	0.88	0.86	151
1	0.76	0.84	0.80	202
2	0.80	0.81	0.80	195
3	0.64	0.79	0.71	183
4	0.90	0.85	0.88	205
5	0.90	0.85	0.87	215
6	0.85	0.77	0.80	193
7	0.90	0.94	0.92	196
8	0.90	0.94	0.92	168
9	0.96	0.93	0.95	211
10	0.90	0.97	0.94	198
11	0.95	0.95	0.95	201
12	0.91	0.78	0.84	202
13	0.97	0.89	0.93	194
14	0.89	0.97	0.93	189
15	0.76	0.98	0.86	202
16	0.85	0.96	0.90	188
17	0.95	0.97	0.96	182
18	0.96	0.77	0.86	159
19	0.92	0.35	0.50	136
accuracy			0.87	3770
macro avg	0.88	0.86	0.86	3770
weighted avg	0.87	0.87	0.86	3770

Predicted Category: sci.space